# Recap: Lecture 21

In Lecture 21, we

- Introduced and classified PDEs
- Introduced initial–boundary value problems
- "Derived" the 1D heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{where} \quad \alpha = \kappa / c\,\rho > 0 \text{ "thermal diffusivity"}$$

| Material | $\alpha$ [$\mathrm{cm}^2\,\mathrm{s}^{-1}$] |
|---|---|
| Gold | 1.27 |
| Iron | $2.3 \times 10^{-1}$ |
| Air | $1.9 \times 10^{-1}$ |
| Quartz | $1.4 \times 10^{-2}$ |
| Wood | $8.2 \times 10^{-4}$ |

# Lecture 22

In Lecture 22, we will:

1. Apply separation of variables to the 1D heat equation + initial + boundary conditions

## Heat equation in 1D: Problem 1

Consider the temperature distribution in a rod of finite length. The rod is initially heated to a prescribed temperature, and then quenched in ice to maintain zero temperature. Assume the longitudinal sides of the rod are insulated, so that heat can only escape through the ends.

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$$u(x,0) = f(x) \quad \text{for} \quad 0 < x < L$$

**Boundary conditions:** Dirichlet BCs

$$u(0,t) = u(L,t) = 0 \quad \text{for} \quad t > 0$$

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4. Apply superposition principle to get a series general solution.
5. Apply initial conditions to find the series coefficients.

# Solving the 1D heat equation using separation of variables

1. Separate variables to turn single PDE into multiple ODEs *which are tied together by a constant of separation* $\lambda$

# Solving the 1D heat equation using separation of variables

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions. *This involves a systematic analysis of three possible cases for the constant of separation.*

# Solving the 1D heat equation using separation of variables

3. Solve the remaining ODEs for each eigenvalue.

# Solving the 1D heat equation using separation of variables

4. Apply Superposition Principle to get a series general solution. *The series will have unknown coefficients to be determined.*

5. Apply the initial conditions to find the series coefficients.

# Solving the 1D heat equation using separation of variables

Let $L = \pi$ and

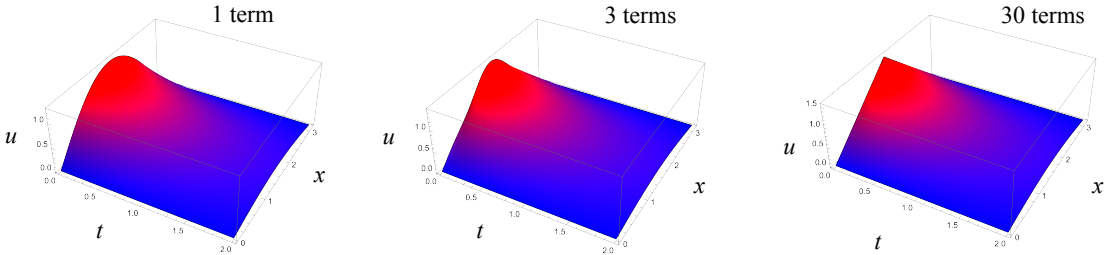
$$f(x) = \begin{cases} x & 0 \leqslant x < \frac{L}{2} \\ L - x & \frac{L}{2} \leqslant x \leqslant L \end{cases}$$

# Solving the 1D heat equation using separation of variables

The heat distribution along the rod is

$$u(x,t) = -\frac{4L}{\pi^2}\sum_{m=1}^{\infty}\frac{(-1)^m}{(2m-1)^2}\mathrm{e}^{-\alpha\omega_{2m-1}^2 t}\sin\omega_{2m-1}x$$

$$\alpha = 1, \quad L = \pi$$




The quartz case ($\alpha = 1\,\mathrm{mm}^2\,\mathrm{s}^{-1}$) and $L = \pi\,\mathrm{mm}$.